# 2025

## COMPUTER SCIENCE AND ENGINEERING

**Paper : CSC-1002**

**(Advanced Software Engineering)**

**Full Marks : 70**

*The figures in the margin indicate full marks.*

*Candidates are required to give their answers in their own words as far as practicable.*

Answer ***question no.*** **1**, ***question no.*** **2**, and ***any four*** from the rest.

**1.** Answer ***any five*** of the following : 2×5

(a) Analyze the security implications of using containers.

(b) Explain the key components of a Use Case Diagram.

(c) What are the benefits of using container orchestration platforms like Kubernetes?

(d) What is meant by risk prioritization?

(e) What is the difference between a product and sprint backlog?

(f) What are the impact of DevOps on traditional IT operations?

(g) Who owns software risk?

**2.** Comment on the correctness of the following statements and justify your opinion (***any five***) : 4×5

(a) "Once a baseline is created, it can be changed at any time without restrictions."

(b) "The product backlog in Scrum is fixed and does not change once the project begins."

(c) "Price-to-Win estimation always leads to accurate cost prediction for software projects."

(d) "Sequence diagrams are better than communication diagrams for showing structural organization of object collaborations."

(e) "The main role of a version control system in DevOps is to store code without tracking changes."

(f) "Generalization in Use Case Diagrams is only applicable to actors, not to use cases."

(g) "Configuration audit ensures that all changes made to a baseline comply with configuration status reports."

**3.** (a) List and briefly describe the types of structural UML diagrams.

(b) What is the primary focus of each diagram?

(c) Create and analyze a Use Case Diagram for an online shopping platform, detailing the actors, use cases, and relationships. 2½+2½+5

4. (a) Discuss the concept of Continuous Integration and Continuous Deployment (CI/CD) in DevOps.

   (b) How do these practices enhance software quality and delivery speed? 5+5

5. (a) Explain the risk management steps with an illustration.

   (b) What are the factors affecting software risks?

   (c) Briefly discuss different types of risks for a software project. 4+3+3

6. (a) Explain the concept of containerization.

   (b) Discuss how containers differ from virtual machines.

   (c) What are the advantages they offer for software deployment and scalability? 3+3+4

7. Consider the following inventory management application project : The application has five screens, each with three views, and nine data tables for four servers and five clients. The application generates three reports, each with five sections, from the nine data tables, for three servers and four clients. 15% reuse of object points is expected. The developer's experience is nominal, but the capability in the given environment is high.

   (a) Identify the size metric (Application Points / Object Points) for this project, considering screens, reports, and tables.

   (b) Calculate the New Application Points (NAP) after accounting for reuse.

   (c) Comment on how developer capability influences productivity in the Application Composition Model.

   (d) Comment on the main difference between COCOMO and COCOMO-II model. 2+1+2+5

8. (a) Illustrate a scenario where implementing DevOps practices led to significant improvements in a software project.

   (b) Discuss the challenges faced and how they were overcome. 6+4